

SAGNIK KUMAR NATH

B.Tech Electrical Engineering (2027) | Renewable Energy & Power Systems

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PROFILE

Final-year Electrical Engineering student (CGPA **8.76**, graduating 2027) seeking a **Graduate Engineer Trainee** role in renewable energy (commissioning, O&M and performance of wind and solar assets). Grounded in generators, machines, protection and distribution; hands-on testing and calibration exposure in NABL-accredited Govt. of India labs and safety documentation at ONGC. Open to relocation and site postings.

EDUCATION

B.Tech, Electrical Engineering – Institute of Engineering & Management (IEM), Kolkata CGPA **8.76** · 2023–27
Class XII (CBSE, Science) – Aditya Academy Senior Secondary · 82.8% 2022
Class X (WBBSE) – Bongaon High School · 90.71% (Mathematics 100) 2020
Relevant coursework: Power Systems · Electrical Machines · Switchgear & Protection · Control Systems · Basic Electrical & Electronics · Industrial Automation I & II

EXPERIENCE

Engineering Intern, Electrical Laboratory – National Test House (ER), Govt. of India · Kolkata Sept 2026
• Rotated through NABL-accredited Electro-Technical Calibration, Thermal Calibration and Electrical Testing laboratories (1–30 Sept 2026).
• Observed test and calibration procedures for electrical instruments and equipment against national standards, including EV and battery-cell testing at the Alipore facility.
• Built working familiarity with measurement accuracy, traceability and test documentation.
Industrial Trainee – Oil and Natural Gas Corporation (ONGC) · MBA Basin, Kolkata May – June 2026
• Reviewed DGMS and OISD safety and regulatory documentation; identified and flagged citation inconsistencies.
• Prepared compliance reports and coordinated with cross-functional engineering teams.

PROJECTS

Rooftop Solar PV System Design (2.75 kWp, grid-connected)
• Sized the array and system, estimated ~3,900 kWh/year using PVGIS, and assessed shading, irradiation and payback.
Battery-Constrained MPC for Adaptive Cruise Control in EVs — paper in preparation (IEMRE, Springer LNEE)
• Designing a Model Predictive Controller with battery-degradation constraints and Kalman-filter state estimation; modelled and simulated in MATLAB, with a web-based simulation platform.
Other builds: real-time anaesthesia monitoring system (sensor-based vitals and alerts) · Wi-Fi controlled mecanum-wheel robot

TECHNICAL SKILLS

Power & electrical: power system fundamentals (per-unit, load flow and fault-analysis concepts), transformers, machines and motor control, switchgear and protection (relays, CT/PT, breakers), generation and distribution systems, earthing, electrical testing and calibration practice
Renewables: solar PV system sizing, energy-yield and payback analysis (PVGIS), grid-connected systems, net metering; generator and power-plant electrical fundamentals for wind and solar assets
Tools: MATLAB / Simulink, AutoCAD (basic 2D drafting), MS Excel / Word / PowerPoint, LaTeX (Overleaf), Git

CERTIFICATIONS & ACHIEVEMENTS

• **L&T EduTech** (Coursera, Sept 2026): Basics of Electrical Protection System · Power Plant Electrical Systems & Generator Fundamentals · Electrical Power Generation – An Industrial Outlook · Electrical Power Distribution
• Motors and Motor Control Circuits – University of Colorado Boulder (Coursera) · AutoCAD for Beginners: Drafting Blueprints & Schematics (Coursera project)
• NPTEL – Solar Photovoltaic System Design, IIT Roorkee (exam Oct 2026) · Solar Energy Basics – SUNY (Coursera) · CAEV Program – JSW MG Motor / Skill-Lync
• Participant, ISRO Bharatiya Antariksh Hackathon 2025 · GATE 2027 (Electrical Engineering) – registered · IEEE Student Member & IEEE IAS member; co-organised IEEE SBC IEM's Aarambh 2.0 (Aug 2026)
Languages: English, Bengali, Hindi | **Availability:** full-time from 2027; willing to relocate & work at sites